

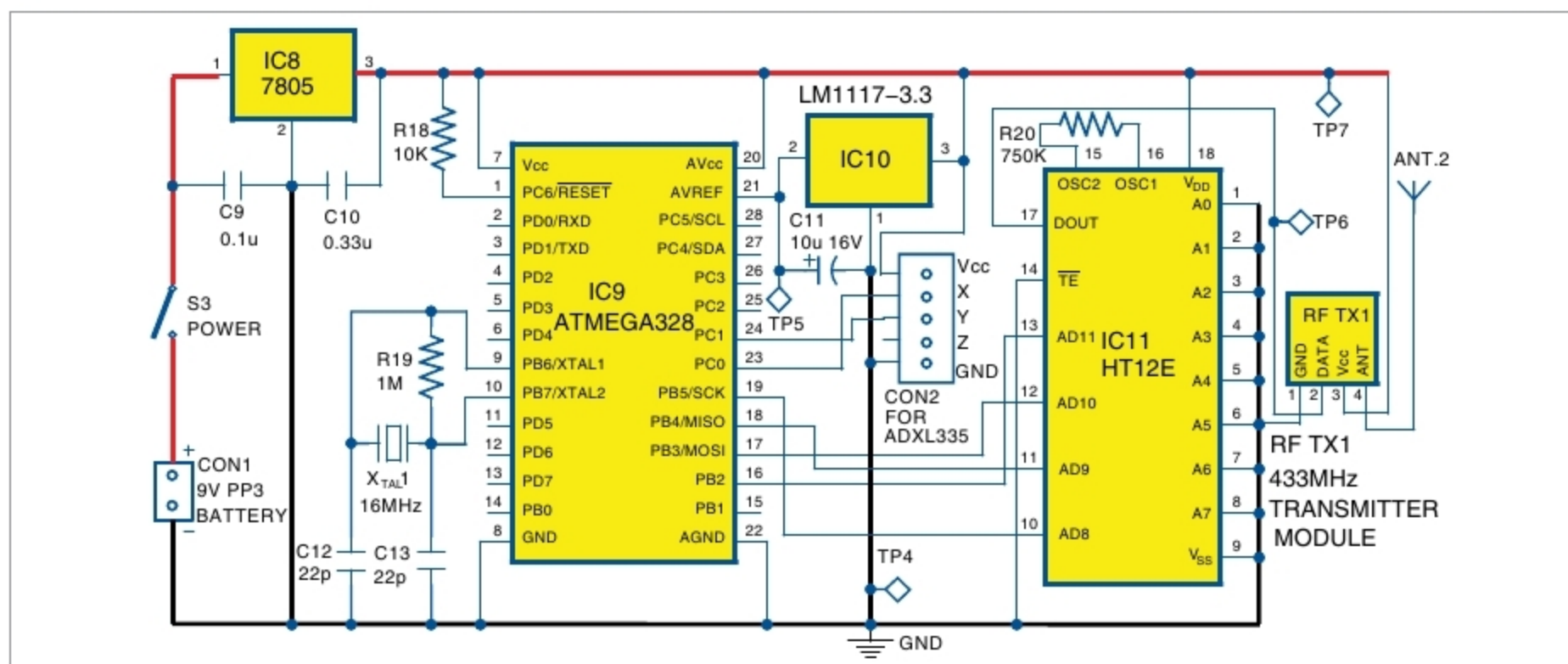


Fig. 2: Circuit diagram of the gesture transmitter unit

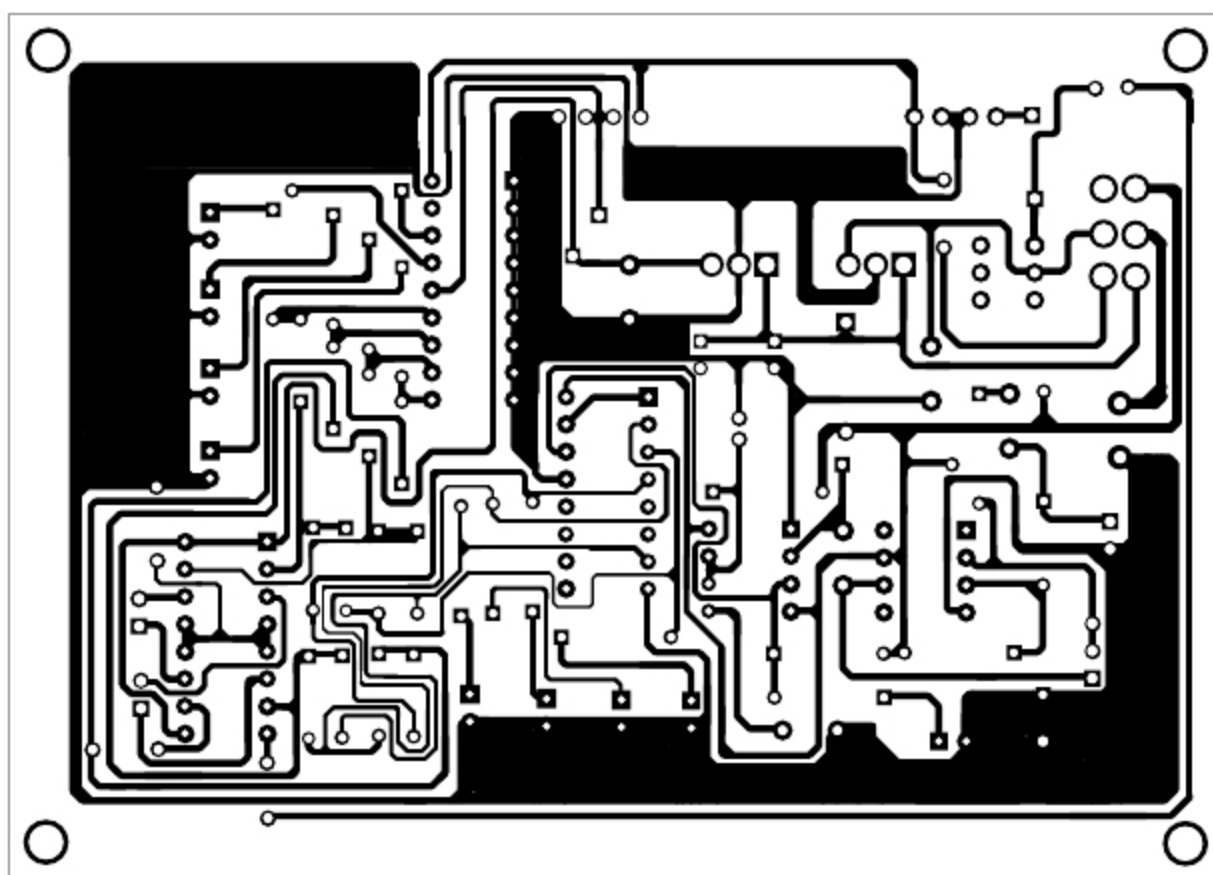


Fig. 3: Actual-size PCB layout of the clap- and gesture-controlled robot (Receiver)

Uno for in-system programming (ISP) given in the IDE, by selecting File→Examples→Arduino ISP. Once the bootloader is uploaded into the MCU, gesture.ino code of this project can be uploaded.

Receiver unit. The receiver unit (shown in Fig. 1) is built around 433MHz receiver module RF RX1, HT12D decoder (IC7) and the motor driver section. DPDT switch (S1) is used for changing mode of operation from clap to gesture and vice versa.

ADXL335 sensor data transmitted through RF TX1 is received at the receiver unit through RF RX1. It

goes to decoder HT12D and then to L293D motor driver section, including LED6 through LED9 and diodes D10 through D13. LEDs are used to indicate the signal status on various output pins of the decoder (IC7). Motor driver section is common for both modes (clap and gesture) for this project.

Switch S1 plays an important role here. With this double-pole double-throw (DPDT) switch, you can change mode of operation to clap or gesture control. When S1 is in clap mode, only clap section will be on and active. Similarly, when it is

in gesture mode, only gesture circuit will be on and active. Switch S2 along with signal diodes (D2 through D13) are used to control the motors (M1 and M2).

Construction and testing

An actual-size PCB layout of the clap- and gesture-controlled robot is shown in Fig. 3 and its components layout in Fig. 4. PCB layout for the transmitter unit is shown in Fig. 5 and its components layout in Fig. 6. Assemble the components as per the circuit diagrams on the PCBs for gesture transmitter and robot control circuit, separately.

After assembling, enclose the robot control section in a suitable box. Mount switches (S1 and S2) on the panel. Make sure that the antenna (ANT.1) and mic (MIC1) are exposed by providing suitable holes for them on the box. Connect 12V battery (BATT.1) and motors (M1 and M2), and place BATT.1 on the chassis of the robot.

Enclose the transmitter section also in a suitable box. Make a small hole each for the antenna (ANT.2) and on/off switch (S3) on the panel. A 9V PP3 battery may be used as power supply for the transmitter unit. Connect the 9V battery to CON1.

For testing, keep S1 in clap mode and switch on S2. Clap in front of the mic. Using a screwdriver, gently turn the wiper of preset VR1 left or right to change sensitivity as per require-